

Assignment 2– Command Line Arguments

Description:

This assignment is to write a C program that allocate memory for structure personalInfo and assign values and write personalInfo. Then, managing a buffer to store strings.

Approach / What I Did:

First thing I did is to allocate memory for a structure called personalInfo using malloc. After that, I assign my arguments into appropriate part of personalInfo and write personalInfo using writePersonInfo function. And, I freed memory that I use for personalInfo. Also, I allocate memory for buffer to store strings.

Issues and Resolutions:

My first issue was to understand what hexadecimal number and their significance.

I figure out this by printing out every values, such as every Ascii codes for individual letters in my name, integers in hexadecimal and binary formats. This approach helped me to understand the concept of hexadecimal representation.

My second issue was that I have difficulty storing string in the buffer while ensuring that there was no wasted space.

I resolved it by adding offset to the pointer. This involved copying memory from a specific point in the buffer by adding an offset to the pointer.

Analysis:

```
0xfffffec562317
0xfffffec562320
37009317
44081f
111110000000100000010001
3
```

First line is address of the firstname.

Second line is address of lastname.

Third line is my student ID in hexadecimal.

Fourth line is languages in hexadecimal.

Fifth line is languages in binary number format in reversed order.
The last line is enum level in hexadecimal.

```
000000: 17 23 56 EC FF FF 00 00 20 23 56 EC FF FF 00 00 | .#V???.. #V???..
000010: 17 93 00 37 03 00 00 00 1F 08 44 00 46 6F 75 72 | .?.7.....D.Four
000020: 20 73 63 6F 72 65 20 61 6E 64 20 73 65 76 65 6E | score and seven
000030: 20 79 65 61 72 73 20 61 67 6F 20 6F 75 72 20 66 | years ago our f
000040: 61 74 68 65 72 73 20 62 72 6F 75 67 68 74 20 66 | athers brought f
000050: 6F 72 74 68 20 6F 6E 20 74 68 69 73 20 63 6F 6E | orth on this con
000060: 74 69 6E 65 6E 74 2C 20 61 20 6E 65 77 20 6E 61 | tinent, a new na
000070: 74 69 6F 6E 2C 20 63 6F 6E 63 65 69 76 65 64 20 | tion, conceived
```

Based on the first Photo, bytes at address 000000 - 000007 highlighted yellow represent the address where pointer, firstname, points.

Bytes at address 000008-00000f highlighted red represent the address where pointer, lastname, points.

Bytes at address 000010-000013 highlighted blue represent the my student ID in hexadecimal format.

Bytes at address 000014-000017 highlighted green represent the enum grade level in hexadecimal format.

Bytes at address 000018-00001b highlighted purple represent the language in hexadecimal format

Based on languages in binary format, matched languages are C, Java, JavaScript, Python, C++, HTML, Mips Assembler, and R which are exactly what I typed in my code.

Bytes at address 00001c-00007f not highlighted represent the message that every letter in Ascii code format.

Screen shot of compilation:

```
parallels@ubuntu-linux-22-04-02-desktop:~/Desktop/csc415-assignment2-bufferandst
ruct-Pangwoo$ make
gcc -c -o Lee_Gwangwoo_HW2_main.o Lee_Gwangwoo_HW2_main.c -g -I.
gcc -o Lee_Gwangwoo_HW2_main Lee_Gwangwoo_HW2_main.o assignment2M1.o -g -I.
```

Screen shot(s) of the execution of the program:

```
----- CHECK -----  
Running the check for Gwangwoo Lee  
Name check is 0 by 0  
Student ID: 922784535, Grade Level: Senior  
Languages: 4458527 (44081F)  
Message:  
Four score and seven years ago our fathers brought forth on this continent, a ne  
w nation, conceived  
  
The Check Succeeded (0, 0)  
  
END-OF-ASSIGNMENT  
000000: 17 23 E8 DC FF FF 00 00 20 23 E8 DC FF FF 00 00 | .#????.. #????..  
000010: 17 93 00 37 03 00 00 00 1F 08 44 00 46 6F 75 72 | .?.7.....D.Four  
000020: 20 73 63 6F 72 65 20 61 6E 64 20 73 65 76 65 6E | score and seven  
000030: 20 79 65 61 72 73 20 61 67 6F 20 6F 75 72 20 66 | years ago our f  
000040: 61 74 68 65 72 73 20 62 72 6F 75 67 68 74 20 66 | athers brought f  
000050: 6F 72 74 68 20 6F 6E 20 74 68 69 73 20 63 6F 6E | orth on this con  
000060: 74 69 6E 65 6E 74 2C 20 61 20 6E 65 77 20 6E 61 | tinent, a new na  
000070: 74 69 6F 6E 2C 20 63 6F 6E 63 65 69 76 65 64 20 | tion, conceived
```